Sri Lanka Institute of Information Technology

Year 2 Semester 1

IT2140 Database Design and Development

Assignment 01

Web-based pet care system-DDD_Jan_I_31

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Requirement Specification

1. System Purpose

The **Web-Based Pet Care System** is designed to automate and streamline the daily operations of a pet care center.
It enables customers to register, manage their pets, and book services online.
Staff can view and update booking statuses, while administrators manage users, services, and generate reports.
The system reduces manual paperwork, minimizes booking errors, and improves customer experience.

2. Users of the System

User Type	Role Description
Customer	Registers, logs in, adds/edits pet profiles, browses services, creates bookings, views booking history.
Staff	Views daily bookings, checks pet details, updates booking status (confirmed, completed, cancelled).
Admin	Manages customers, staff, services, generates reports (e.g., revenue, popular services, staff workload).

3. Functional Requirements

ID	Requirements
FR1	Customers can register and log in securely Admin can view, edit, or delete customer and staff accounts
FR2	Customers can add, update, view, and delete pet profiles
FR3	Customers can view all available services (name, price, duration)
FR4	Customers can create a booking by selecting a pet, service, staff, date, and time
FR5	Staff can view all bookings assigned to them

FR6	Staff can update booking status (Pending ,Confirmed ,Completed / Cancelled)
FR7	Admin can view, edit, or delete customer and staff accounts.
FR8	Admin can add, update, or remove services
FR9	Admin can generate reports (e.g., monthly revenue, most booked service, pet type popularity)
FR10	System prevents double-booking the same staff at the same date/time

4. Non-Functional Requirements

ID	Requirement	Quality Attribute
NFR1	Pages load within 3 seconds under normal use	Performance
NFR2	System supports at least 50 concurrent users without slowdown	Scalability
NFR3	Passwords are hashed; role-based access control (RBAC) enforced	Security
NFR4	Data is backed up daily; recovery possible within 1 hour	Reliability
NFR5	Interface is simple, consistent, and requires no external training	Usability
NFR6	System works on Chrome, Firefox, and Edge (responsive design).	Compatibility
NFR7	99.5% uptime (excluding planned maintenance)	Availability

5. Constraints

ID	Constraints
C1	Must be a web-based system (not mobile app or desktop software)
C2	Built using standard web technologies (e.g., HTML, CSS, JavaScript, PHP/Node.js, MySQL)
C3	Must run on a university-provided server or localhost for demonstration
C4	No external payment gateway integration in this version
C5	All data must be stored in a relational database (MySQL)
C6	Only authenticated users can access the system (no public booking without login)

Evidence of Requirement Gathering (Bot Interaction)

Figure: Bot response about system purpose

Project Discussion Bot - Year2 > Hi I'm planinng to drawing EER diagram

Auto

April 10

Hi I'm planinng to drawing EER diagram

Project Discussion Bot - Year2

Hi 😊
That's good. I can help you step by step.

Before we start, I need to know about your project.
Please tell me, in simple words:

What is your project title?
Who will use this system?
Is it a system, website, or mobile app?

Just answer these. Then we continue.

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2:24 PM

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Auto

1.web-based pet care system

Project Discussion Bot - Year2

Good 👍
Web-based pet care system. I understand.

Now tell me two more things, please.

Who will use this system?
For example: pet owners, vets, pet shops, admins.

What does the system mainly do?
For example: book appointments, manage pets, buy products, etc.

Just explain in simple words.

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1.web-based pet care system 2. customer, center staff, Administrators 3. web site

Project Discussion Bot - Year2

Good. I understand your project now.

This project is for a Sri Lankan pet care company called **Happy Paws Care Lanka**.
They provide pet care services using a website for customers and staff.
The company wants to improve service and manage work easily.

Here are **5 client personas** for your project:

1. **Ms. Anushka Perera – Operations Manager**
2. **Mr. Sahan Fernando – Pet Care Center Staff Supervisor**
3. **Ms. Dilini Jayawardena – Customer Support Officer**
4. **Mr. Kasun Silva – System Administrator**
5. **Mrs. Nadeesha Kumari – Pet Owner (Customer)**

You can now ask questions like you are talking to a real client.

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Project Discussion Bot - Year2 > Hi I'm planinng to drawing EER diagram

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Hi I'm MS.Anushka perera ,can you explain what the system should i do

Project Discussion Bot - Year2

Hello, I'm **Ms. Anushka Perera**, Operations Manager at **Happy Paws Care Lanka**.
Our company wants a simple website to manage all pet care services in one place. This system should help us work faster and give better service to customers.

The system should allow customers to register, add their pets, and request services like grooming or medical care.
Center staff should see bookings, update service status, and manage daily work.
Administrators should control users, services, and view reports to run the business smoothly.

You can now ask me about what information we need to keep, or what problems we face now.

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Auto

1st i want to know website first page how to create

Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

The first page of the website is very important. It is the page customers see first. It should look clean, friendly, and easy to understand. When someone opens the site, they should quickly know this is a pet care website.

On the first page, we want a simple welcome message. It should say who we are and what we do. For example, we care for pets and offer services like grooming and health care. We also want nice pictures of pets to make people feel happy and safe.

The first page should also help users go to other parts of the website. Customers should easily find buttons like register, login, services, and contact us. Everything should be clear and not confusing.

If you want, next you can ask me what information should appear on the first page, or what problems customers face when visiting our website now.

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Project Discussion Bot - Year2 > Hi I'm planinng to drawing EER diagram

Auto

What info should be on the first page?

Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

The first page should clearly show who we are. It should have the company name and a short welcome message. When a customer opens the site, they must quickly understand that this is a pet care service website.

We should show a short description about what services we offer, like pet grooming, health care, and day care. Friendly pet images are important because they create trust and a good feeling. The page should also show our contact details so customers know how to reach us.

There should be clear buttons to go to other pages. For example, login, register, view services, and contact us. The first page should be simple, clean, and easy to use, especially for first-time visitors.

If you want, you can ask me what problems customers face now, or what message we want to show on the homepage.

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Auto

What message should be on the homepage?

Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

The homepage message should be warm and friendly. It should tell visitors that we care about pets and their safety. It must make pet owners feel they can trust us.

A good message is something like this:
"Welcome to Happy Paws Care Lanka. We provide loving and professional care for your pets. From grooming to health services, we are here to keep your pets happy and healthy."

This message is simple and clear. It explains what we do and shows that we care.
Next, you can ask me how this message helps customers, or what feeling we want customers to have when they see the homepage.

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What feeling should customers get from the homepage?

Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

When customers open the homepage, they should feel **safe and calm**. They must feel that their pets will be cared for with love and responsibility. They should feel **happy and confident** to use our service.

The homepage should also give a **friendly and welcoming feeling**. Customers should feel that our staff are kind and helpful. This feeling will make them stay on the website and trust us.

If you like, next you can ask me what customers should do first after opening the homepage.

how to create my 2nd page

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Auto

how to create my 2nd page

Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

The second page should guide customers after the homepage.
For our website, the best second page is the **Services page**.

On this page, customers should see what services we provide.
For example, pet grooming, health checkups, boarding, or training.
Each service should have a short, simple description so customers understand easily.

This page should help customers decide what they want.
From here, they should feel confident to register or book a service.

If you want, next you can ask me what details should be shown for each service, or what page should come after this.

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Auto

what details should be shown for each service, or what page should come after this.

Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

For each service on the Services page, customers should see clear and simple details.
They need the service name, a short description, and who the service is for.
They should also see the price or starting cost and how long the service takes.
A small image helps customers understand the service better.

After the Services page, the next page should be the **Registration or Login page**.
From there, customers can create an account or log in to book a service.
This helps us know who the customer is and manage their pets and bookings easily.

If you want, you can now ask me what information we need from customers when they register.

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What info is needed for customer registration?

Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

For customer registration, we need only basic and important details.
We should ask for the customer's full name, email address, phone number, and a password. This helps them log in safely and helps us contact them.

We also need their home address, because some services may depend on location.
After registration, they can add pet details later. This keeps registration simple and easy.

If you want, next you can ask me what pet details we need to collect after registration.

What pet details do you want to collect?

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Hello. This is **Ms. Anushka Perera**, Operations Manager.

After the customer registers, we need basic pet details. We should collect the pet's name, type of pet like dog or cat, and the breed. We also need the pet's age and gender.

We should ask about the pet's health information. This includes allergies, medical problems, and vaccination details. This helps our staff give safe and correct care.

We may also ask the pet's size and any special care notes. This helps us prepare before the service.
If you want, you can next ask me how staff will use these pet details in daily work.

How staff use pet details daily?

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Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

Our staff use pet details every day to give correct care. Before a service, they check the pet's age, size, and health notes so they know how to handle the pet safely. This helps avoid problems like allergies or stress.

During the service, staff use these details to choose the right tools, time, and treatment. After the service, they update notes so next time the care will be even better.

If you want, you can ask me how bookings are handled by staff next.

how bookings are handled by staff next.

Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

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Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

When a customer makes a booking on the website, our staff can see it in their system. They check the service type, date, time, and pet details before accepting it. This helps them prepare early and avoid mistakes.

After accepting the booking, staff update the booking status. On the service day, they mark it as in progress and later completed. This way, everyone knows what work is done and what is coming next.

If you want, you can next ask me how administrators use booking information.

How do admins use booking info?

Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

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Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

Administrators use booking information to manage the whole business. They check how many bookings come each day and which services are popular. This helps them plan staff work and service times.

Admins also use booking data to solve problems. If a booking is delayed or cancelled, they can see what happened and take action. They also use this information to create simple reports for management decisions.

If you want, you can next ask me what reports admins want from the system.

How do admins use reports?

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Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

Admins use reports to understand how the business is doing. They look at reports to see how many bookings we get, which services are used most, and which days are busy. This helps them plan staff and improve services.

Reports also help admins make decisions. They can see problems early, control costs, and plan future improvements. This way, the company can grow and give better service to customers.

If you want, you can ask me what types of reports are important for management.

what types of reports are important for management.

Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

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Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager.

For management, the most important reports are simple and clear. We need booking reports to see how many services are done each day and which services are popular. This helps us plan staff and time.

We also need customer reports to understand how many customers we have and how often they use our services. This helps us improve customer satisfaction. Another important one is service performance reports, so we can see which services are doing well and which need improvement.

These reports help management make good decisions and improve the business.
If you want, you can next ask me how these reports help in future planning, or you can say "next" to talk to another person.

Next to talk to another person

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Sign in

Chat

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Project Discussion Bot - Year2

Okay. Now you are talking to **Mr. Sahan Fernando**, Pet Care Center Staff Supervisor.

Our company Happy Paws Care Lanka runs daily pet care services using this website. I manage staff work and daily bookings at the center. I focus on smooth service and avoiding mistakes.

You can ask me about staff work, daily problems, or how the system helps staff.

staff work, daily problems, or how the system helps staff.

Project Discussion Bot - Year2

Hello, I'm **Mr. Sahan Fernando**, Pet Care Center Staff Supervisor.

Staff work is very busy every day. We handle many pets, bookings, and customers at the same time. The main daily problems are missed bookings, wrong pet details, and confusion about service times.

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Next to talk to another person

Project Discussion Bot - Year2

Okay. Now you are talking to **Ms. Dilini Jayawardena**, Customer Support Officer.

I talk with customers every day. I answer their questions and help them use the website. I care about customer happiness and easy use.

You can ask me about customer problems, questions they ask, or how the website helps customers.

customer problems, questions they ask, or how the website helps customers.

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Project Discussion Bot - Year2

Hello, I'm **Ms. Dilini Jayawardena**, Customer Support Officer.

Customers often feel confused at first. They ask how to register, how to add their pets, and how to book a service. Some worry about prices and if their pet will be safe.

The website helps them a lot. Everything is clear and in one place. They can see services, book easily, and check their booking status without calling us.

If you want, you can say **"next"** to talk to another person, or ask me more about customer needs.

Next to talk to another person

Project Discussion Bot - Year2

Okay. Now you are talking to **Mr. Kasun Silva**, System Administrator.

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Project Discussion Bot - Year2

Okay. Now you are talking to **Mr. Kasun Silva**, System Administrator.

I manage the website and user accounts at Happy Paws Care Lanka.
I make sure the system is safe, works well, and data is correct.

You can ask me about admin work, system control, or security.

admin work, system control, or security.

Project Discussion Bot - Year2

Hello, I'm **Mr. Kasun Silva**, System Administrator.

My admin work is to manage users and the system every day. I create staff and admin accounts, remove inactive users, and make sure all data is correct. I also watch the system to see if anything is wrong.

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Discussion Bot - Year2 > Hi I'm planinng to drawing EER diagram Auto

admin work, system control, or security.

Project Discussion Bot - Year2

Hello, I'm Mr. Kasun Silva, System Administrator.

My admin work is to manage users and the system every day. I create staff and admin accounts, remove inactive users, and make sure all data is correct. I also watch the system to see if anything is wrong.

For system control, I check that the website is working well. I manage services, prices, and basic settings. If there is an error or problem, I fix it quickly so work does not stop.

For security, I protect customer and pet information. Only the right people can see or change data. This helps keep the system safe and builds customer trust.

If you want, you can ask me what problems admins face, or say "**next**" to finish or talk to someone else.

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Project Discussion Bot - Year2 > Hi I'm planinng to drawing EER diagram

Auto

next

Project Discussion Bot - Year2

Okay 😊
Now we have finished talking to all the main people in the company.

You have collected good **system requirements** from: customers, staff, support, and admin.

Next step for you is to **list the information (data)** the system must store.
This will help you later to draw the **EER diagram**.

For example, you can think about: customers, pets, services, bookings, staff, and reports.

When you are ready, you can ask me like this:
"What main data items should I list for my system?"
I will help you step by step. 👍

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Auto

What main data items should I list for my system

Project Discussion Bot - Year2

Good question 👍
This is the **right step before drawing your EER diagram**.

For your web-based pet care system, you should list the **main data items** the system needs to store. These are the real-world things the business works with every day.

First, you need **Customer information**.
This includes customer name, contact details, address, login details, and account status. Without this, customers cannot use the system.

Next, you need **Pet information**.
Each pet belongs to a customer. You need to store pet name, pet type, breed, age, gender, health details, and special care notes.

Then, you need **Service information**.
This includes service name, description, price, duration, and availability. This helps customers choose services

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Then, you need **Service information**.
This includes service name, description, price, duration, and availability. This helps customers choose services and helps staff plan work.

You also need **Booking information**.
This connects customers, pets, and services. It stores booking date, time, service type, booking status, and assigned staff. This is very important for daily work.

Next is **Staff information**.
This includes staff name, role, contact details, and work status. Staff use the system to manage bookings and services.

You also need **Administrator information**.
Admins manage the system. Their data includes admin name, login details, and permissions.

Finally, you may have **Report information**.
This is used by admins and management. Reports are based on bookings, services, and customers to help make decisions.

These are the **main data items** you should list.
From these, you can now identify entities and relationships for your EER diagram.

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Auto

These are the **main data items** you should list.
From these, you can now identify entities and relationships for your EER diagram.

When you are ready, you can ask:
"How do these data items connect to each other?"

How do these data items connect to each other?

Project Discussion Bot - Year2

Good question 👍
I will explain this in **very simple words**, like a real discussion.

Think of the system like daily work at our pet care center.

A **customer** uses the website first.
One customer can have **many pets**.

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Auto

Think of the system like daily work at our pet care center.

A **customer** uses the website first.
One customer can have **many pets**.
But each pet belongs to **only one customer**. This is very important.

A **customer** makes a **booking**.
A booking is always linked to **one customer** and **one pet**.
Without a customer and a pet, a booking cannot happen.

Each **booking** is for **one service**.
For example, grooming or health check.
But the same service can be used in **many bookings** by different customers.

When a booking is made, **staff** handle it.
One staff member can handle **many bookings** in a day.
But each booking is handled by **one staff member** at a time.

Admins do not make bookings.
They manage customers, staff, services, and bookings.
They also use booking data to create **reports**.

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Customers own pets.
Customers make bookings for pets.
Bookings use services.
Staff handle bookings.
Admins control everything and see reports.

This is how your data items connect.
From this, you can clearly draw relationships in your EER diagram.

If you want, next you can ask:
"What are the main relationships I should draw in my EER diagram?"

What are the main relationships I should draw in my EER diagram?

Project Discussion Bot - Year2

Hello. This is **Ms. Anushka Perera**, Operations Manager

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